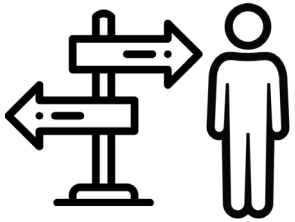
# *MGMT 3130*
# *Judgment and Decision Making in Organizations*

## *Unit 2.7*
## *Escalation of Commitment*



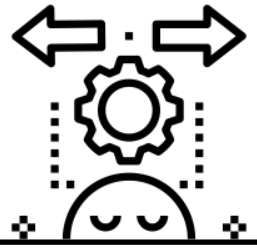
# *Where are we now?*

## Part 1: Decision Analysis

- Unit 1.1 Defining your decision problems
- Unit 1.2 Decisions involving multiple objectives
- Unit 1.3 Decision making under uncertainty

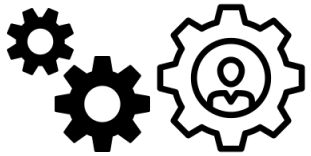
## Part 2: Common Traps of Human Irrationality

- Unit 2.1 Two systems of thinking
- Unit 2.2 Availability heuristic
- Unit 2.3 Representativeness heuristic
- Unit 2.4 Bounded awareness



- Unit 2.5 Prospect theory
- Unit 2.6 Planning fallacy
- **Unit 2.7 Escalation of commitment**
- Unit 2.8 Temporal decision making
- Unit 2.9 Biases in group decision making

## Part 3: JDM Issues in Contemporary Organizations



- Presentation day 1: Unethical behavior
- Presentation day 2: Going green
- Presentation day 3: Diversity and inclusion
- Presentation day 4: Performance appraisal

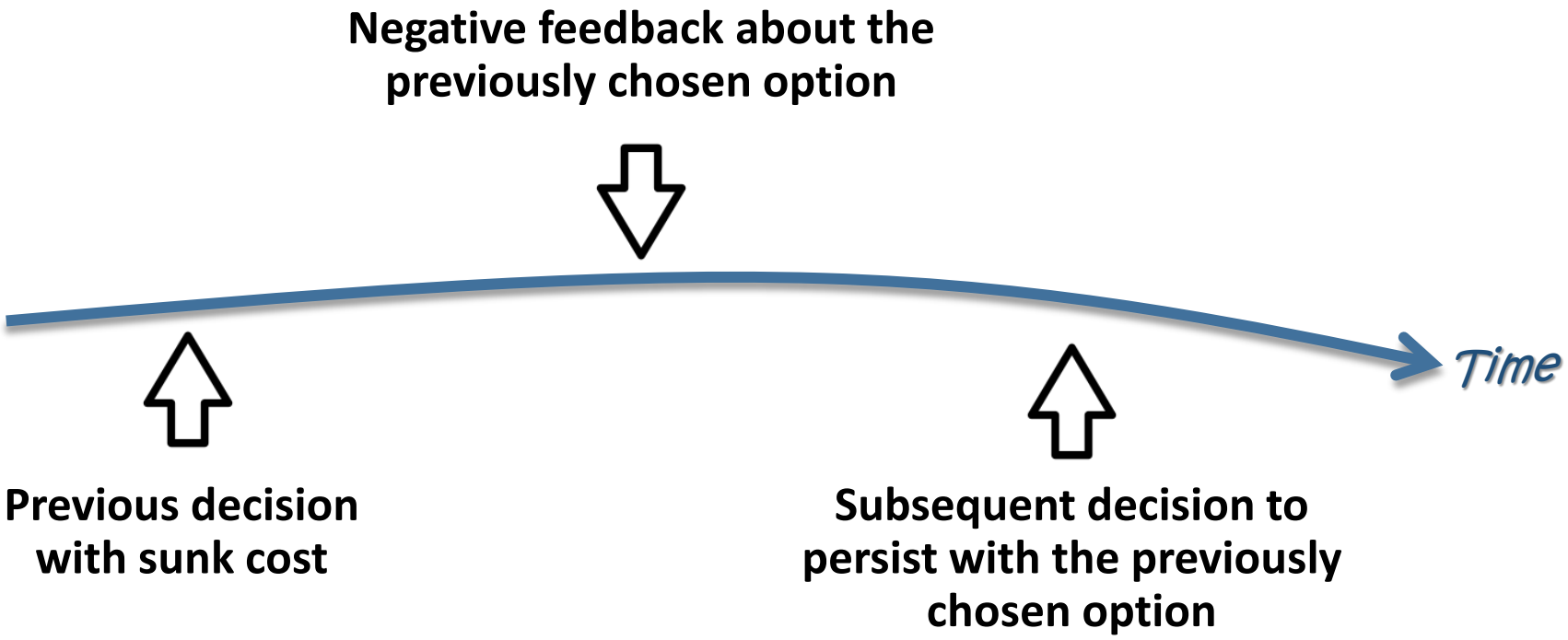
## Unit 2.7 Escalation of Commitment

- What's the escalation of commitment?
- Mechanisms underlying the escalation of commitment
  - The expectancy explanation
  - Sunk cost fallacy
  - Goal substitution effect
  - Self-justification theory
  - Self-presentation theory
  - Psychological ownership and confirmation bias
- Debiasing strategies for the escalation of commitment

# ESCALATION OF COMMITMENT

***Escalation of commitment*** happens when decision makers continue to dedicate resources (e.g., time and money) to a failing course of action.

- Escalation of commitment has the following features:



Examples of the escalation of commitment:

Stock price dropped

Time

Bought a stock

Bought more of the stock at low price

Found himself not interested in being a dentist

Time

Started a dental degree

Graduated and got a dissatisfying dentist job

Spent a lot of $$ but no R&D achievement

Time

A company started a R&D project

Invested even more $$

Your turn:

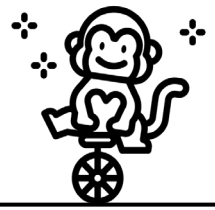
## MECHANISMS UNDERLYING THE ESCALATION OF COMMITMENT

Sometimes it makes sense to persist in face of negative feedback (i.e., it's rational):

- The expectancy explanation

Sometimes the decision to escalate our commitment is based on irrational thoughts:

- Sunk cost fallacy
- Goal substitution effect
- Self-justification theory
- Self-presentation theory
- Psychological ownership and confirmation bias



## *The Expectancy Explanation*

***The expectancy explanation***: Decision makers exhibit greater escalation tendency when the expected value of persistence (vs. withdrawal) is larger, e.g.,

- When the value of goal attainment is higher
- When the probability of goal attainment is higher

Consider the following study:

- Participants were asked to play the role of an assistant director of the World Bank, and to choose among 3 potential locations for developing an industrial complex in Kenya.
- 1st decision was made by participants.
- Negative feedback: $70 million of the original allocation of $80 million had been spent for the project, yet the industrial complex was only 50% completed.

Staw, B.M., & Ross, J. (1978). Commitment to a policy decision: A multi-theoretical perspective. *Administrative Science Quarterly*, 23(1), 40-64.

- Two experimental conditions:

*Temporary setback condition:*

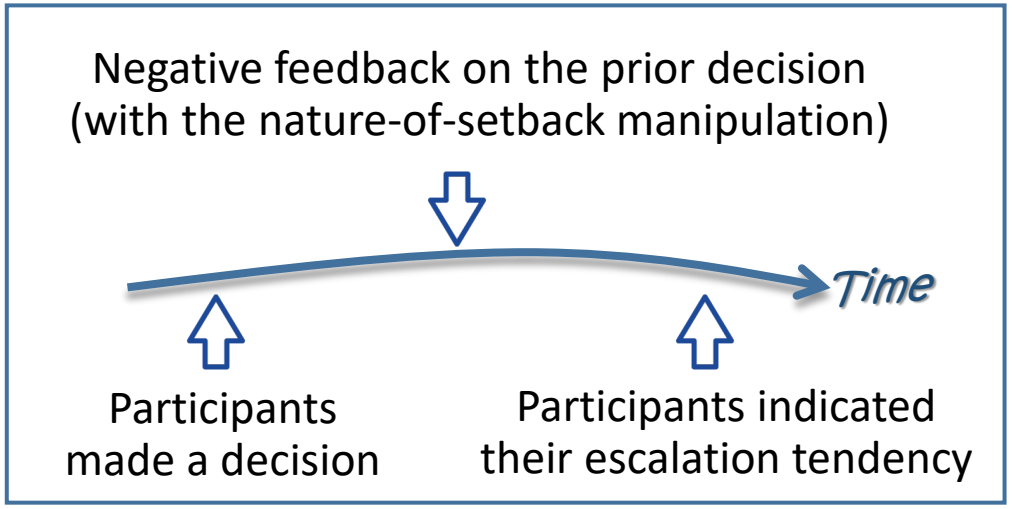
Highly unusual amount of rain throughout the region was responsible for the delay in the project but it was unlikely to constitute a recurring setback.

*Enduring setback condition:*

The slow progress was caused by enduring setback. There was corruption in the area and that raw materials for the project had been diverted.

- All participants made a 2nd decision: "An additional $70 million has been made available to the World Bank. How much additional funding ($0-70M) do you wish to allocate to this Kenya project?"

- Results:
  Participants invested more into the Kenya project when the setback is of temporary (vs. enduring) nature [i.e., when the probability of goal attainment is higher].



## *Sunk Cost Fallacy*

*A demonstration*

- ☞ Scenario 1: Imagine that you paid HK$100 for a ticket to see a horror movie. Halfway through the movie you realize the movie is extremely boring, and all the scenes look very fake. How likely you will now walk out of the movie theater?
- ☞ Scenario 2: Imagine that you received a free ticket (which is worth HK$100) to see a horror movie. Halfway through the movie you realize the movie is extremely boring, and all the scenes look very fake. How likely you will now walk out of the movie theater?

- Results:

| | **Mean likelihood to leave halfway through the movie*** |
|---|---|
| Scenario 1 | |
| Scenario 2 | |

| **You paid** | **If you close the account now, the account balance will be…** |
|---|---|
| $100 | Negative, "in RED" |
| Nothing | Zero |

* Scale: 0 (not at all likely) <--------> 10 (extremely likely)

Results of MGMT 3130 in-class activity

In the previous case, the money used to purchase the movie ticket is a *sunk cost* – a cost that was incurred in the past and cannot be recovered once spent.

Rationally (from standard economics sense), sunk costs should be ignored when making decisions about future investment. All that matters should be *future costs and benefits*.

In the previous case, what should you consider?

- The enjoyment you get from doing something else if you walk out of the movie theater.
- The enjoyment (or disappointment) of watching the rest of the movie.
- Not the sunk cost!

In the context of the escalation of commitment, decision makers are unwilling to withdraw from the failing course of action because they are *unwilling to accept the previously committed resources as a loss* (closing an account "in red" is extremely painful).

Consider an experiment:

Condition 1:

As the president of an airline company, *you have spent $9 million into the R&D of a radar-blank plane*. Now you find that another firm begins marketing a radar-blank plane, and apparently their plane is much faster and far more economical than yours.

Will you invest next $1 million to the radar-blank plane project?

Condition 2:

As the president of an airline company, *you have to decide whether or not to invest $1 million into a R&D project – to build a radar-blank plane.* Now you find that another firm begins marketing a radar-blank plane, and apparently their plane is much faster and far more economical than yours.

Will you invest $1 million to the radar-blank plane project?

Results:

*85% of respondents said "Yes"*

*17% of respondents said "Yes"*

Arkes, H.R., & Blumer, C. (1985). The psychology of sunk cost. *Organizational Behavior and Human Decision Processes*, 35(1), 124-140. Experiment 3.

## *The Tennessee-Tombigbee Waterway Project*

- In the Tennessee-Tombigbee Waterway Project, the original cost estimate in 1970 was $323 million. Difficulties were encountered during construction. The actual cost upon completion in 1985 were just under $2 billion, and the benefit/cost ratio is far less than projected.
- At the Congressional review on 4th Nov 1981, Senator Sasser mentioned: *"Terminating the [Tennessee-Tombigbee Waterway] project at this late stage of development would represent a serious waste of funds already invested."*

Cited in Arkes, H.R., & Blumer, C. (1985). The psychology of sunk cost. *Organizational Behavior and Human Decision Processes*, 35(1), 124-140.

## *Goal Substitution Effect*

***Goal substitution effect***: As decision makers approach project completion, they substitute a *project completion goal* for their *original project success goal*. This goal substitution leads to the escalation of commitment towards the project.

- For example: Consider a failing R&D project, although the original goal may have been profit maximization, as decision makers approach project completion, the new goal may be nothing more than completing the R&D project regardless of financial returns.

Consider an experiment:

- Participants were told that a R&D project is now either *10% completed* (in the "*low* degree of completion" condition) or *90% completed* (in the "*high* degree of completion" condition).

  [Note: Sunk cost was held constant across the two conditions.]
- Participants learned about negative feedback on the R&D project, and were asked to allocate additional fund to the project.
- Results: Escalation tendency is higher in the high (vs. low) degree of completion condition.

Conlon, D. E., & Garland, H. (1993). The role of project completion information in resource allocation decisions. *Academy of Management Journal*, 36(2), 402-413. Experiment 2.

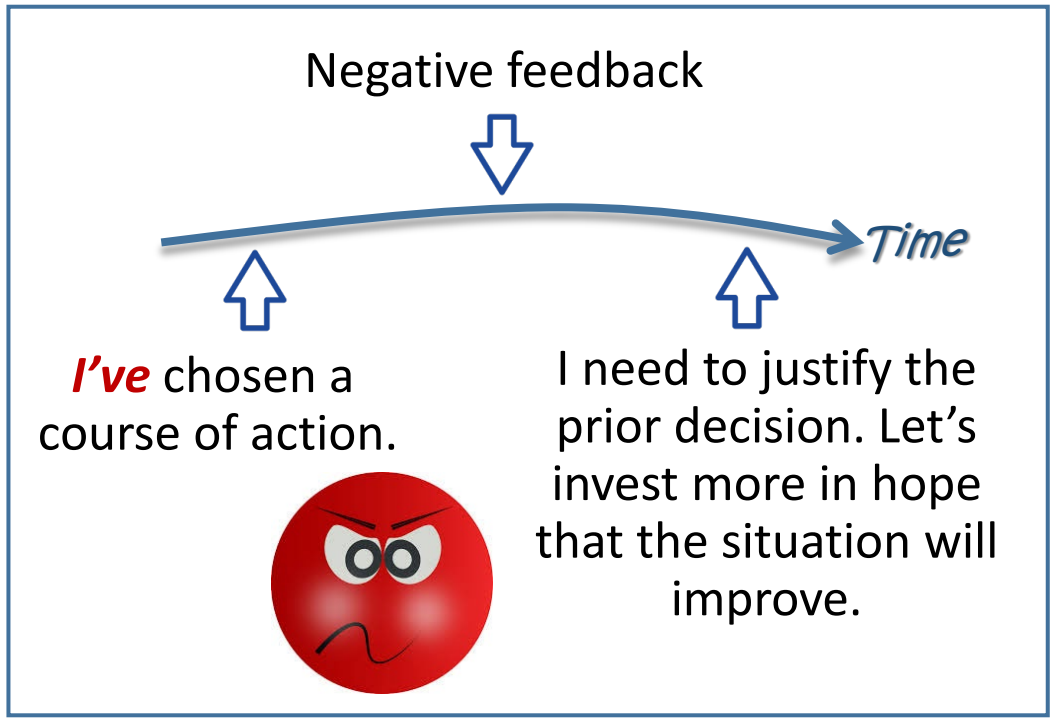
## *The 1996 Mount Everest disaster*

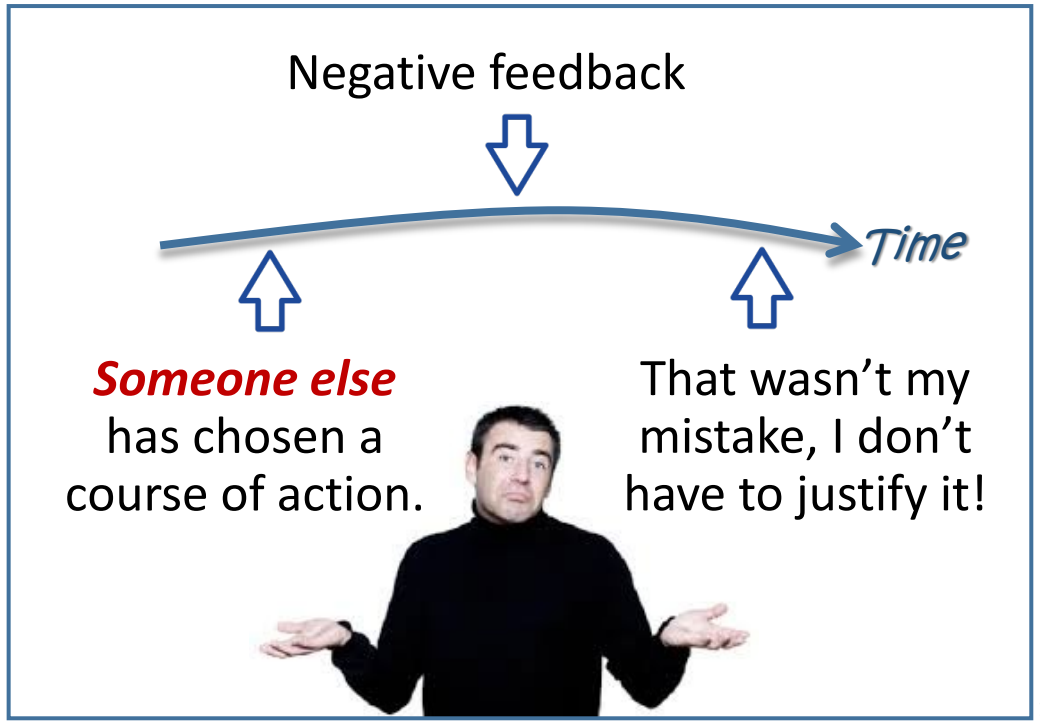
- The final leg of the climb is called the Death Zone. Climbers need to exert maximum physical effort for over 12 hours to scale the peak and then return before a storm, avalanche, exhaustion, and/or oxygen deprivation strands them on the mountain.
- The summit fever:
  *"I have to reach the summit this time!" Climbers risk everything to achieve the completion goal.*
- In 1996, some climbers failed to quit despite that they knew a storm was coming. They ended up staying on the mountain forever.

## *Self-justification Theory*

***Self-justification theory***: Negative feedback calls the validity of the prior decision into question. Decision makers don't like to think that their past decision was a mistake. To maintain their ego, they invest more to the failing course of action in hope to turn the situation around.

*<u>The effect of personal responsibility on escalation of commitment</u>*



Consider an experiment:

- Participants were told to act as the Financial Vice President of a company that has two divisions: *Consumer Products* and *Industrial Products*.

| *High responsibility condition* | *Low responsibility condition* |
| --- | --- |
| ⇨ 1st decision: Select which division to receive R&D funding.<br>⇨ Learned that the selected division has performed poorly.<br>⇨ 2nd decision: Allocate additional $20 million among the two divisions. | ⇨ Learned that a preceding financial officer has allocated R&D funding to a division.<br>⇨ Learned that the funded division has performed poorly.<br>⇨ Decision: Allocate additional $20 million among the two divisions. |

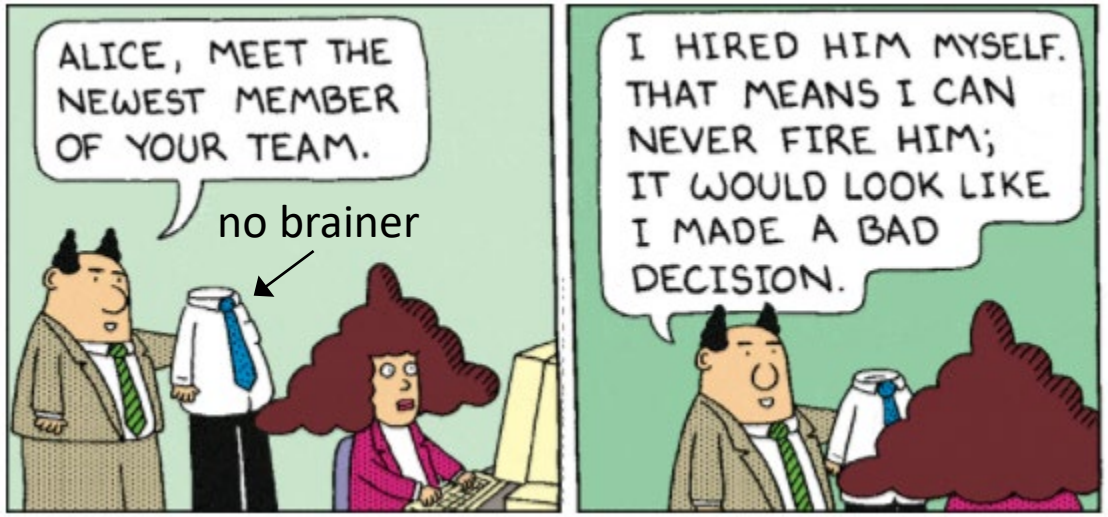
- Results: Participants allocated more $$ to the previously funded division when they were responsible (vs. not responsible) for the previous decision.

Staw, B.M. (1976). Knee-deep in the big muddy: A study of escalating commitment to a chosen course of action. *Organizational Behavior and Human Performance*, 16, 27-44.

## *Self-presentation Theory*

According to ***self-presentation theory***, people are motivated to strategically manage the impressions others have of them (those others could include supervisors, coworkers, followers, etc.).

In escalation situations, decision makers want to avoid any public embarrassment in being associated with a failed project. Moreover, because of the *social norm that values decision consistency*, people try to be seen as good decision makers by persisting with their original choice.




*Source of picture: https://dilbert.com/strip/1998-10-20*

## *Social norm that encourages decision consistency*

"A President must, plainly, show himself to be a man made confident by the courage of his own clear convictions. … The American people find it easy to forgive a leader's great mistakes, but not long meanderings."

--- Fortune Magazine (1978), in an article criticizing President Jimmy Carter

A study has shown that leaders who are consistent in their actions are perceived to be better leaders than those who switch from one line of behavior to another.

In order to be seen as effective leaders, decision makers exhibit the escalation of commitment.

Staw, B.M., & Ross, J. (1980). Commitment in an experimenting society: A study of the attribution of leadership from administrative scenarios. *Journal of Applied Psychology*, 65(3), 249-260.

## *Psychological Ownership and Confirmation Bias*

- ***Psychological ownership***: Decision makers may develop psychological ownership with a chosen course of action, experience a need to cultivate and protect the chosen course of action.

  Psychological ownership can be developed even without personal responsibility due to the psychological connection among individuals.

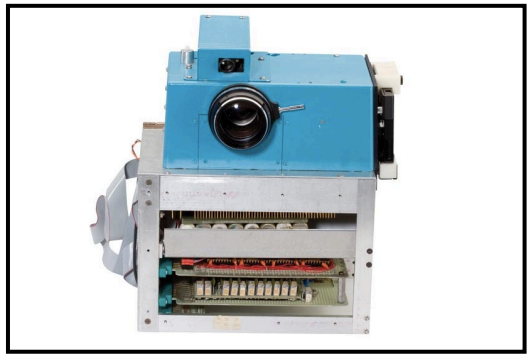
- ***Confirmation bias***: With an aim to protect the chosen course of action (due to psychological ownership), decision makers are likely to notice evidence that supports the prior decision, and ignore negative feedback.

  Being unaware of the negative feedback, decision makers persist with the prior decision.

## *The case of Kodak digital camera*

Kodak Brownie camera (1900)

Kodak Kodachrome color film (1935)



Kodak invented digital camera (1975)

- George Eastman (Kodak's founder) aims to make photography "as convenient as the pencil". In 1900, the Brownie camera brought photography within financial reach of the general public. In 1935, Kodak launched the Kodachrome color film and became the market leader in film business. In 1975, Kodak invented the world's first digital camera; but it held back from manufacturing digital cameras for the mass market, *for the fear of killing its film business*.
- Many Kodak employees are proud of the success of the film business, i.e., *psychologically connected to Kodak's film business*.

- In 1981, a team inside Kodak assessed the threat that would be posed by digital technology. The report concluded that:

  *"The quality of prints from electronic images will not be generally acceptable to consumers as replacement for prints based on the science of photography [i.e., film]."*

  *"The consumer's desire to handle, display, and distribute prints cannot be replaced by electronic display devices."*

  *"Electronic systems (camera and viewing input device for TV) will not be low enough in price to have widespread appeal."*

  There's a whiff of *confirmation bias* in these conclusions. Kodak cannot see the negative feedback.

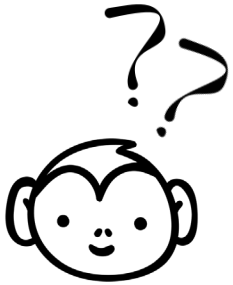
Carroll, P. B., & Mui, C. (2008). *Billion Dollar Lessons: What You Can Learn From the Most Inexcusable Business Failures of the Last Twenty-five Years*. New York: Portfolio.

## DEBIASING THE ESCALATION OF COMMITMENT

Discussion:

Suppose you're a bank loan officer, and you've authorized a loan ($1M) to a small business 2 years ago. The repayment deadline is approaching, but the small business has not repaid the loan yet, and it is now applying for extra loan (another $1M) from you.

Given the mechanisms of irrational escalation of commitment (e.g., sunk cost fallacy, self-justification and self-presentation theories, psychological ownership and confirmation bias), what can you / the bank do to reduce irrational commitment to problem loans?



***Let's examine some examples of debiasing strategies:***

## *1. Remove initial decision makers from subsequent decisions.*

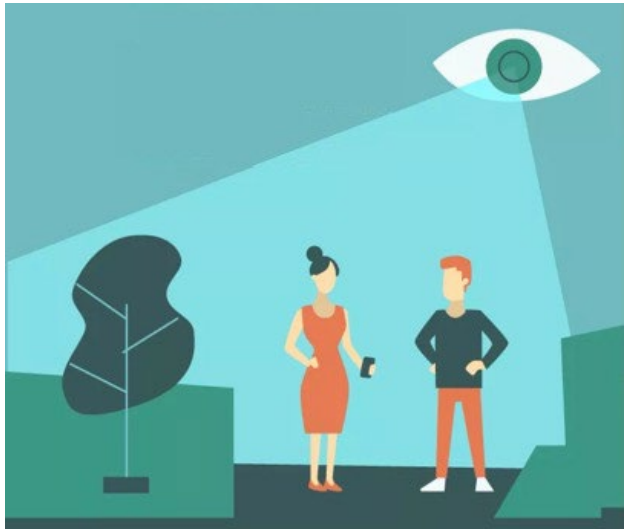
- To reduce irrational escalation, organizations could change the decision maker; the new decision maker is not responsible for the failing course of action, thus has a lower level of justification pressure.
- Data collected from 132 California banks over a 9-year period showed that the *turnover of bank managers led to reduced commitment to problem loans*, manifested as:
  - Increase in the provisions for loan losses ($$ set aside in anticipation of uncollectible loans)
  - Increase in the write-off of bad loans (considered loans as uncollectible)

Staw, B.M., Barsade, S.G., & Koput, K.W. (1997). Escalation at the credit window: A longitudinal study of bank executives' recognition and write-off of problem loans. *Journal of Applied Psychology*, *82*(1), 130-142.

## *2. Use self-distancing (i.e., to take the perspective of a neutral observer).*

- To reduce irrational escalation, decision makers could do *self-distancing* and *adopt an observer's perspective* – in their mental image they should see themselves as part of the field of view.

  By taking an observer's perspective, decision makers *disregard the emotional attachment to the past*, *engage in more balanced reasoning*.



- Example: Intel abandoned a once-promising line of business – memory chips:

  Intel's business had been built on memory chips. The business was profitable until Japanese competitors came into market. In 1978, Intel experienced a steep decline in its market share for memory. In 1981, IBM selected Intel's microprocessor to power the IBM personal computers.

During early 1980s, Intel became a company with two products: memory (which is under threat) and microprocessor (which seems to have opportunity to grow). There was a debate about resource allocation among these two businesses.

In 1985, after months of fruitless debate, Andy Grove (President of Intel) had an insight during a meeting with Gordon Moore (CEO) …

> I looked out the window at the Ferris Wheel of Great America amusement park revolving in the distance, then I turned back to Gordon and I asked, "if we got kicked out and the board brought in a new CEO, what do you think he would do?" Gordon answered without hesitation, "He would get us out of memories."
>
> I stared at him, numb, then said, "Why shouldn't you and I walk out the door, come back in, and do it ourselves?"

Using *self-distancing*, executives at Intel prevented irrational escalation, exited its original memory business to concentrate on microprocessors.

Grove, A. S. (1996). *Only the Paranoid Survive*. New York: Currency Doubleday.

## *Intended Learning Outcomes For Unit 2.7*

By now, you should be able to:

- Define the escalation of commitment, and discuss its impacts in organizational context and in your personal life.
- Describe the mechanisms underlying the escalation of commitment, and then suggest ways to debias.

## *Time For Reflection*

Recall an experience in which you've exhibited irrational escalation of commitment. How would you prevent it from happening again in the future?